

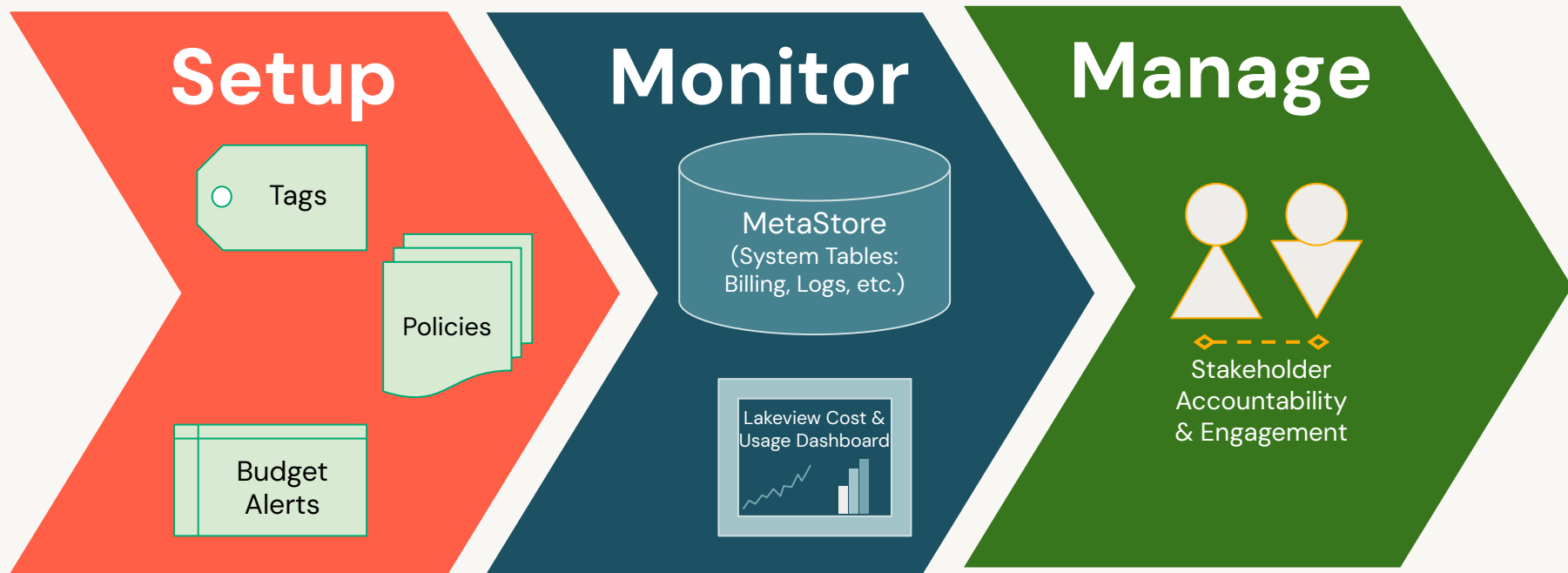


Cost Management Best Practices



Critical components of Cost Management

Suite of features enabling cost management within the Databricks Platform

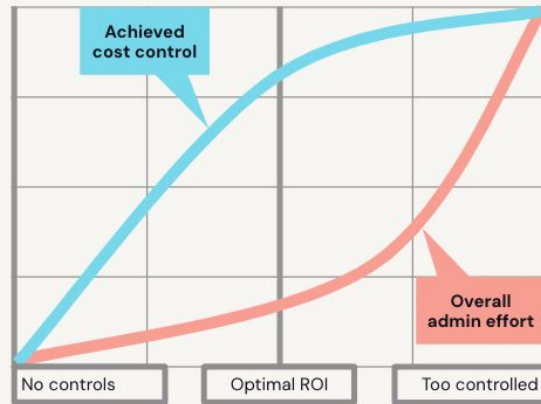


Efficient Enterprise Cost Controls

Build guard rails that scale

- Foundational controls scale well, provide flexibility, and make key usage insights available
- Controls take effort to define, monitor, and iterate
- Excessive controls can restrict the flexibility and initiatives of new project teams
 - Requests for changes to controls consume platform team time
 - Platform teams often become blockers to innovation

Consider both effort and return when designing new controls



Jobs-to-Be-Done for Cost Management



Access



Organize

Setup



Report

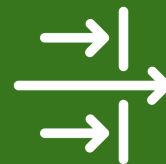


Optimize

Monitor



Forecast



Control

Manage

Jobs-to-Be-Done for Cost Management



Access



Organize

Setup



Report

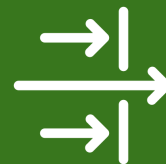


Optimize

Monitor



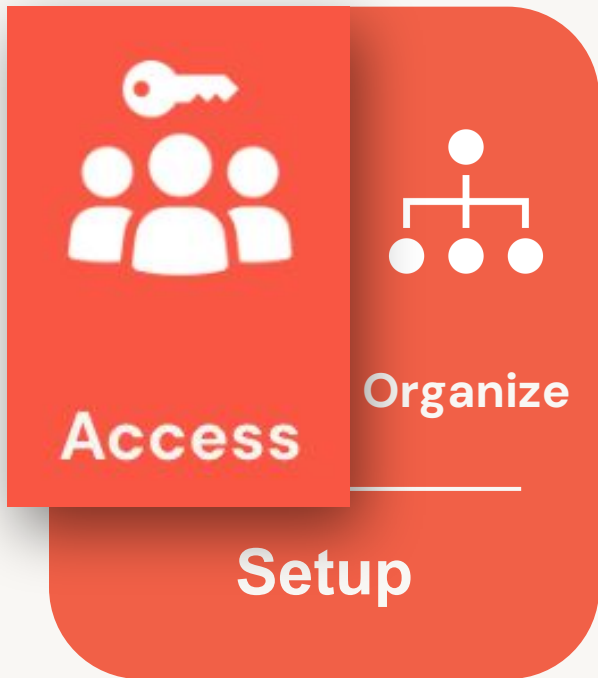
Forecast



Control

Manage

Jobs-to-Be-Done for Cost Management



Setup requires appropriate permissions.

A **Platform Admin** should be completing the tasks in this jobs-to-be-done guide and delegate to users/stakeholders were outlined in the guide.

[Additional context on Admin types w/in Databricks](#)

Jobs-to-Be-Done for Cost Management



Access



Organize

Setup

Organizing is the critical step

Here are the core components

1. Develop expected tagging practices & policies
 - a. General tags & if Specific tags are needed
 - b. Serverless Policies to apply tags to serverless usage
2. Ensure Unity Catalog is setup in one or more workspace to have a metastore active. **Note only account Admins case see billing system tables.**
3. Admin implements Budgets via account console

Tagging best practices

Best Practice Components on Tags

What are tags?

Tags are the best way to track what's happening across your databricks environment. They are structured as Key:Value pairings. Eg Business Unit (key): 100,101,102,103 (value)

What Keys are right for you?

It's recommended that everyone apply General Keys, and for organizations that want more granular insights, they should apply select high specificity keys that are right for their organization.

Keys (categories to use)

General:

(Everyone should start w/ these)

- Business Unit
- Project
- *(optional)* Environment (dev/test/prod)

High Specificity:

Builder (SI partner name, Built on, etc.), Version, Role, Product, Service, Customer, Other / Custom



- ☐ Tags can be applied [on clusters, SQL Warehouses, and on Jobs](#) (Check [account console usage dash for tag coverage data](#))
- ☐ For applying tags to serverless usage leverage Policies ([link to guide](#)) (Also available: PublicPreview Feature*)

*The PuPr mentioned here only apply for Serverless Notebook/ Jobs, DLT

Applying Tags

Users should apply tags to the resources they create

Clusters

Compute > New compute > UI preview Send feedback

New Cluster

Tags

Business Unit

Add tags

Key Value

Automatically added tags

Vendor	Databricks
Creator	bradford.chang@databricks.com
ClusterName	New Cluster
ClusterId	<Generated after creation>

Advanced options

SQL Warehouses

SQL Warehouses >

UC Metric Tree Dashboard

Overview Connection details

Status Running

Name UC Metric Tree Dashboard (ID: 42ae148a1f2a1839)

Type Serverless

Cluster size 4X-Large

Auto stop After 60 minutes of inactivity

Scaling Cluster count: Active 5 Min 1 Max 12

Channel Current (v 2024.35)

Tags

Budget	opex.eng.data
Service	Metric Tree

Created by Amir Issael

Jobs

Workflows > Jobs >

New Job 2024-10-04 14:50:55

Run Task

Unnamed task

Job details

Job ID 704814786088124

Creator Chris Frenzel

Run as Chris Frenzel

Tags Add tag

Unregistered Add description

Schedules & Triggers

Job parameters

Job notifications

Note to apply tags to serverless usage leverage a Policy (covered in next slide)

Serverless Compute Policies

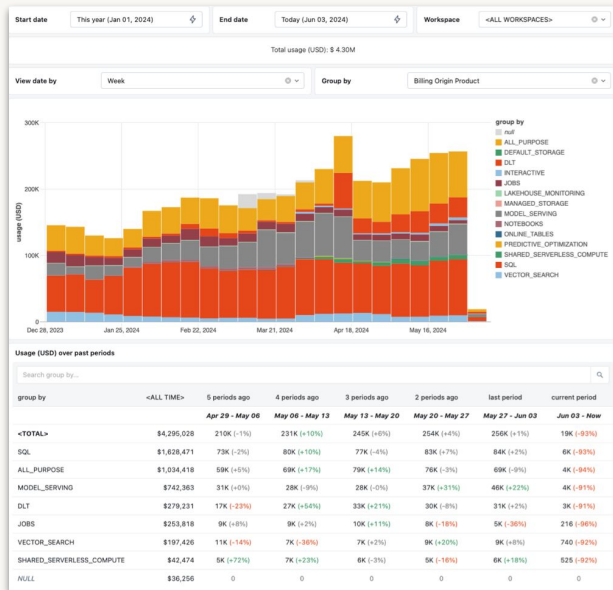
Tagging and Cost Management of Serverless Compute

- With Serverless compute (notebook and job) attribution at the cluster level is not applicable
- However, costs of every single notebook/job run can be attributed by leveraging Budget Policies (in [Public Preview Feature](#)) which automatically add tags to serverless usage
- Policy Setup Guide: [link](#)
 - Consider developing policies that make attributes required or have a default selections. (via [Unlimited Policy](#))
 - Consider: [Admin best practices to cost mgmt](#) for
 - Tagging
 - Auto-scaling
 - auto-termination of serverless compute

Checking account's Tags and Coverage

How do I see what tags are used & associated DBUs?

- Via account console see usage insights and group usage by custom tags ([link](#))
- Session at DAIS covering tag management and Audit (start at 14:50 in [video](#))



How to Set Up Budgets

- In Account Console go to Usage & click Budgets(Preview) tab.
- **Create a new budget, being sure to add the relevant Tags** (recommend including at least Business Unit and Project keys.)

Detailed guide

databricks Databricks Staging is rad

Workspaces
Catalog
Usage
User management
Cloud resources
Previews
Settings

Usage

Consumption **Budgets** Preview

Q Search

Name	Scope	Budget
Proeject B	ALL	\$ 100,000
Project A	ALL	\$ 1,000.00
mate budget	6051921418418893 +1	\$ 120.00
Proeject B	ALL	\$ 100,000
Project C	ALL	\$ 1,000,000

Project C Under Budget

BUDGET
\$1,000,000.00
monthly

Budget Progress

Definitions
ALL

Usage > Budget >

Edit budget

General

Name
A name for this budget.

my team budget

Amount
Specify the monthly amount in US dollars. The amount should reflect the list price of the features you use, including platform add-ons.

\$ 100000

Definitions

Define the scope of the budget. Usage must match all provided workspaces and tag filters.

Workspaces

Workspaces to be included in this budget. Leave this field empty to apply the budget to all current and future workspaces.

billing-workspace-mt-dev-aws-us-west-2-v2 X

Tags

Add tags if you want to limit the budget to usage that includes specific custom tags. Tags are case-sensitive and should be entered exactly as they appear in your usage data. To include multiple values for a key, enter values as a comma-separated list.

Key	Values
Example: teamName	Example: finance, dataScience, sales
key1	
key2	abcd

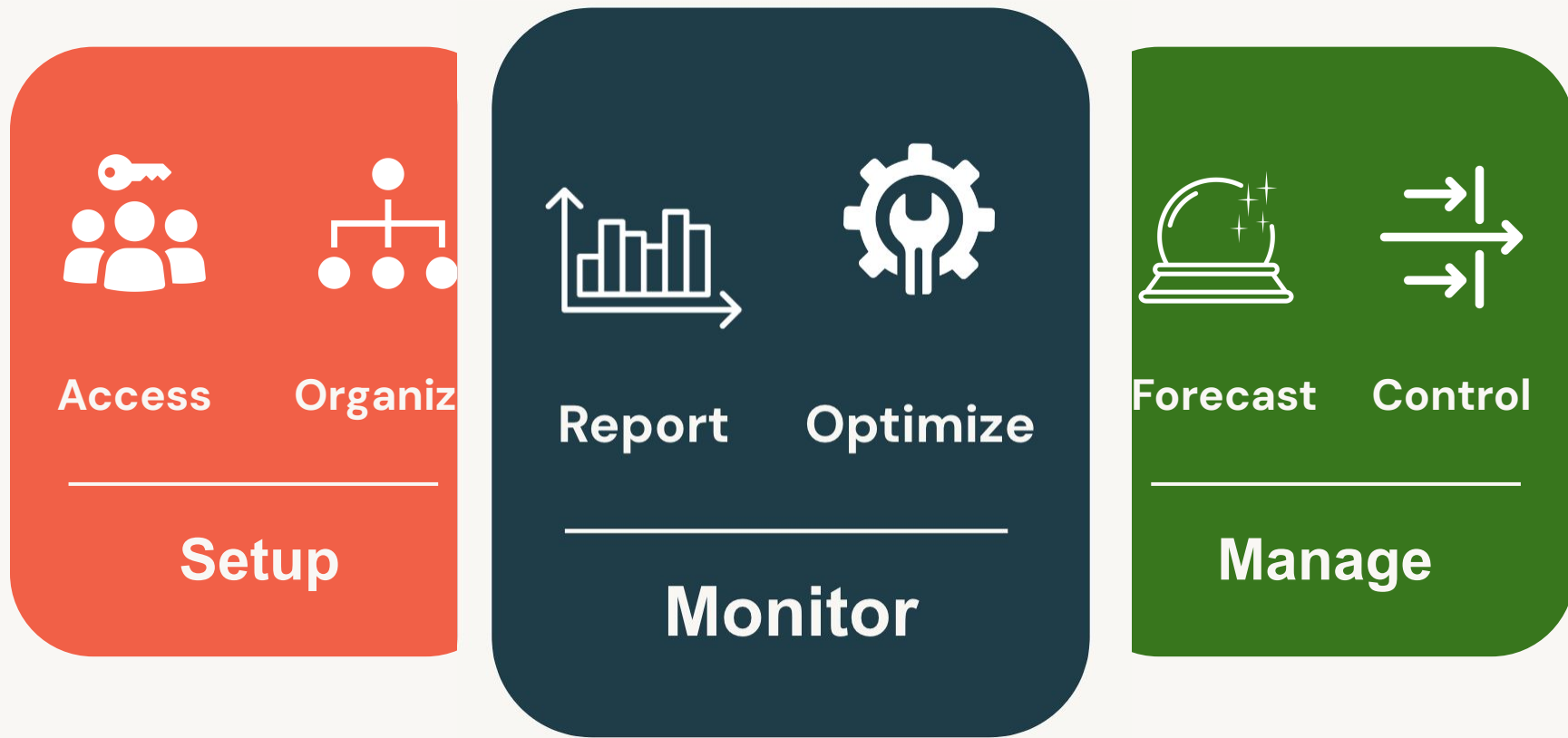
Alerts

Email addresses
A comma-separated list of emails that receive a notification when cumulative usage for this budget exceeds the amount in a given month. There could be up to a 24-hour delay between usage occurring and a notification being sent.

The email notification includes the budget name and tag filters. Keep this in mind when naming the budget, selecting tags, and providing email addresses.

foo@databricks.com

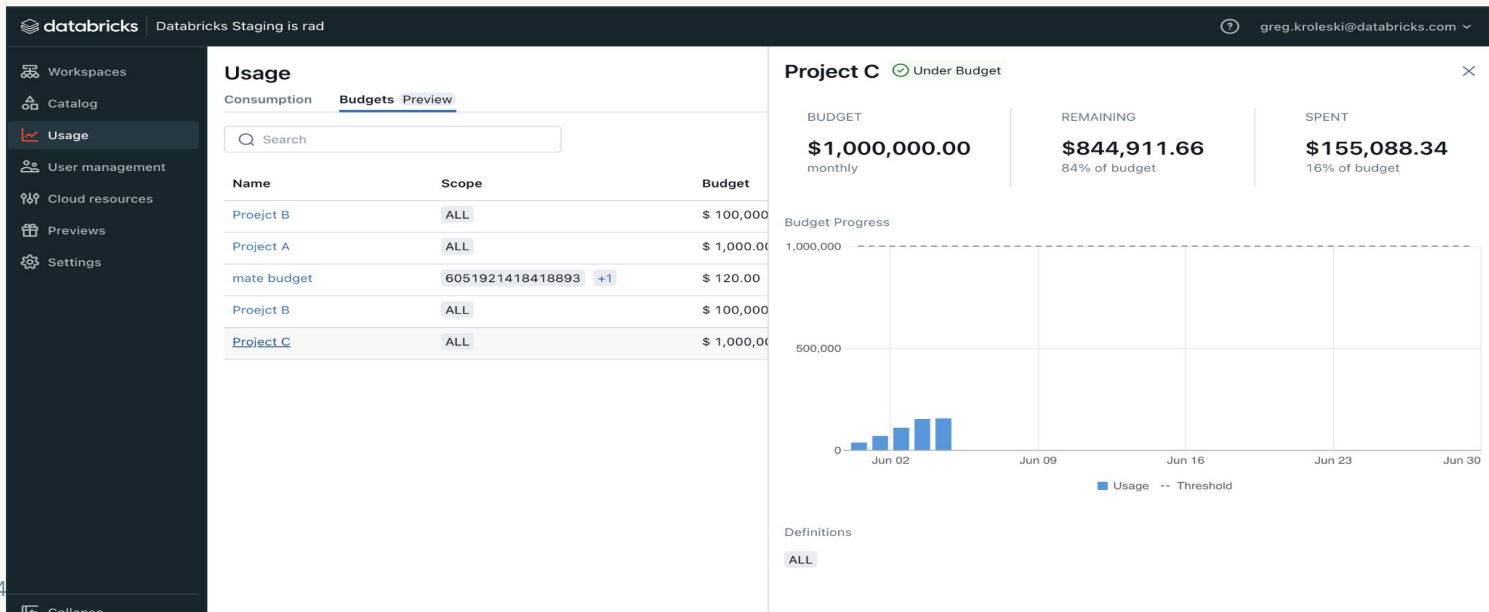
Jobs-to-Be-Done for Cost Management



Budget Monitoring / Reporting

Detailed guide

As an Admin, budgets can be reviewed in account console and shared via alerts function on Budgets



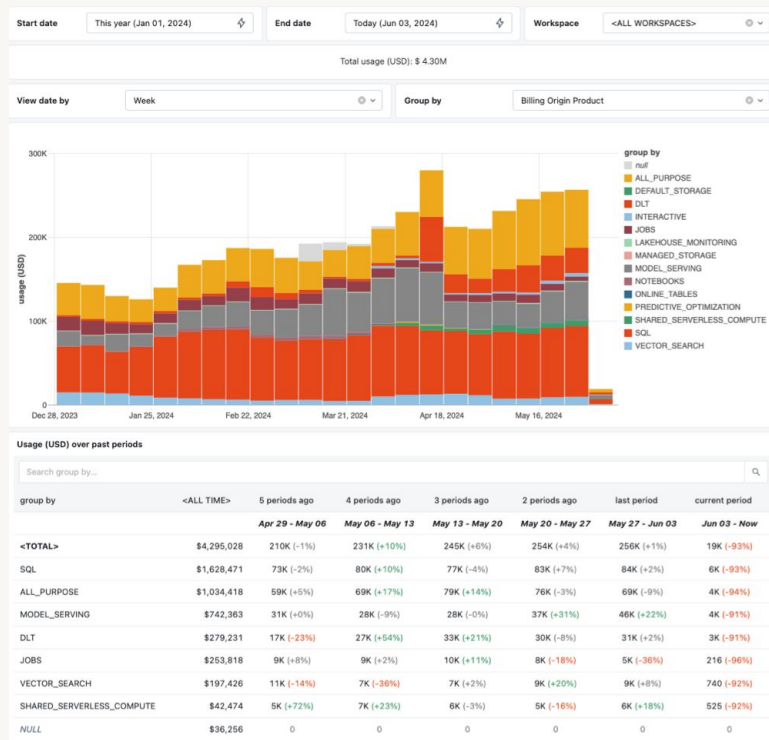
Monitoring Costs & Tagging (Report)

Steps to set-up Lakeview Cost and Usage Dashboard in account console

Usage monitoring in account console:

- Import cost management dashboard into your unity catalog enabled workspace
- Explore usage by: workspace, SKU, tags

Detailed Guide



Jobs-to-Be-Done for Cost Management



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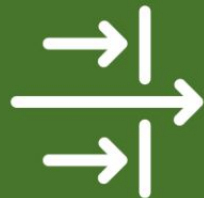


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Managing (Cost Control)

Areas to consider

1. Use Cluster Policies to control which users are able to create clusters, as well as the size and scope of those clusters
2. Design your environment to minimize non-DBU costs generated by Databricks workspaces, such as storage and networking costs
3. Use monitoring tools in order to make sure your expectations of cost are being met and that you have effective practices in place

Review Cost Management Best Practices

